



National University of Computer & Emerging Sciences - NUCES - Karachi

FAST School of Computing

Object-Oriented Programming CL-1004 – LAB 5 TASKS

Task 1: Campus Navigation System

PROBLEM:

Design a **Best-First Search algorithm** to find the most promising route between two locations inside a university campus.

Each location is represented as a node, and heuristic values represent the **estimated walking distance to the destination**.

REQUIREMENTS

- Implement Best-First Search using a priority queue
- Use heuristic values to select the next building
- Maintain OPEN and CLOSED lists
- Stop search once destination building is reached

Task 2: Robot Object Collection Problem

PROBLEM:

A robot is placed in a grid environment containing multiple objects.

Using **Greedy Best-First Search**, the robot must collect the **closest object first** based on heuristic distance.

REQUIREMENTS

- Use Greedy BFS with $f(n) = h(n)$
- Always move toward the nearest object
- Mark collected objects and continue search
- Avoid revisiting cells

Task 3: Emergency Evacuation Route Planning

PROBLEM: Implement an **A*** search algorithm to determine the **fastest evacuation route** in an emergency scenario. The algorithm must minimize **travel time** while avoiding blocked or unsafe routes.

REQUIREMENTS

- Implement A* with $g(n)$, $h(n)$, and $f(n)$
- Update OPEN and CLOSED lists properly
- Ignore unsafe or blocked edges
- Always select node with minimum $f(n)$